

# Lecture 15 - Class Activity ANCOVA

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## Lecture 15: Analysis of Covariance (ANCOVA)

### What is ANCOVA?

ANCOVA (Analysis of Covariance) combines regression and ANOVA to: - Compare group means while adjusting for a continuous covariate - Increase statistical power by reducing residual error - Control for confounding variables

### When to Use ANCOVA

Use ANCOVA when you have: - **Response variable**: Continuous - **Predictor variable**: Categorical (factor/groups) - **Covariate**: Continuous variable that affects the response

### Key Assumptions of ANCOVA

1. **Independence** of observations
2. **Normality** of residuals
3. **Homogeneity of variances** across groups
4. **Linearity** between response and covariate within each group
5. **Homogeneity of slopes** (most critical!) - regression slopes must be equal across all groups

#### ! Critical First Step

Always test for **homogeneity of slopes** before proceeding with ANCOVA. If slopes differ significantly between groups, standard ANCOVA is inappropriate.

## Part 1: Cricket Chirping Analysis

### Data Overview

We want to compare chirping rate of two cricket species: - *Oecanthus exclamationis* - *Oecanthus niveus*

But we measured rates at different temperatures, and there's a relationship between pulse rate and temperature. ANCOVA lets us adjust for temperature effect to get a more powerful test!

```
# Create simulated cricket data based on lecture example
set.seed(456)
n <- 40
species <- rep(c("0. exclamationis", "0. niveus"), each = n/2)
temp <- c(rnorm(n/2, mean = 22, sd = 2), rnorm(n/2, mean = 24, sd = 2))
chirp_rate <- 40 + 2.5 * (temp - 23) + ifelse(species == "0. exclamationis", 10, 0) + rnorm(n,
sd = 3)
cricket_data <- data.frame(species = species, temp = temp, chirp_rate = chirp_rate)

# View data structure

head(cricket_data)
```

```

      species    temp chirp_rate
1 0. exclamationis 19.31296   40.69557
2 0. exclamationis 23.24355   51.78799
3 0. exclamationis 23.60175   50.75553
4 0. exclamationis 19.22222   40.80589
5 0. exclamationis 20.57129   50.16484
6 0. exclamationis 21.35188   46.24225

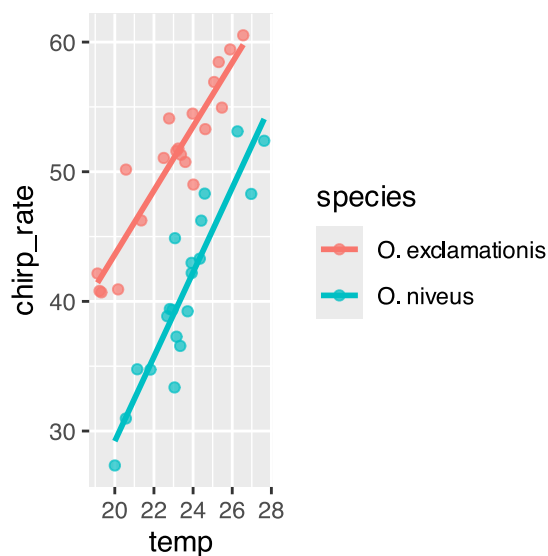
```

```

# Plot with regression lines by species
ggplot(cricket_data, aes(x = temp, y = chirp_rate, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE)

```

`geom\_smooth()` using formula = 'y ~ x'



## Step 1: Test Homogeneity of Slopes

This is the most critical assumption! We test if the regression slopes are equal across all groups.

```

# Test for homogeneity of slopes by including interaction term
cricket_slopes_model <- lm(chirp_rate ~ temp * species, data = cricket_data)
Anova(cricket_slopes_model, type = 3)

```

Anova Table (Type III tests)

Response: chirp\_rate

|              | Sum Sq | Df | F value | Pr(>F)               |
|--------------|--------|----|---------|----------------------|
| (Intercept)  | 6.32   | 1  | 0.9393  | 0.338915             |
| temp         | 620.48 | 1  | 92.1572 | 0.00000000001828 *** |
| species      | 69.76  | 1  | 10.3617 | 0.002724 **          |
| temp:species | 26.08  | 1  | 3.8734  | 0.056796 .           |
| Residuals    | 242.38 | 36 |         |                      |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**Interpretation:** If  $p > 0.05$ , slopes are homogeneous and we can proceed with ANCOVA. If  $p < 0.05$ , slopes differ and standard ANCOVA is inappropriate.

## Step 2: Fit ANCOVA Model

Since slopes are homogeneous ( $p > 0.05$ ), we can fit the ANCOVA model without the interaction term.

```
# Fit ANCOVA model (without interaction)
cricket_ancova <- lm(chirp_rate ~ temp + species, data = cricket_data)

# Get model summary
summary(cricket_ancova)
```

```
Call:
lm(formula = chirp_rate ~ temp + species, data = cricket_data)

Residuals:
    Min       1Q   Median       3Q      Max
-6.0065 -1.9653  0.1923  0.7886  5.9192

Coefficients:
              Estimate Std. Error t value      Pr(>|t|)
(Intercept)   -13.2012     4.7423  -2.784    0.00842 **
temp             2.7926     0.2048  13.634 0.000000000000000530 ***
species0. niveus -11.8005     0.8593 -13.733 0.000000000000000424 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.694 on 37 degrees of freedom
Multiple R-squared:  0.8994,    Adjusted R-squared:  0.894
F-statistic: 165.4 on 2 and 37 DF,  p-value: < 0.00000000000000022
```

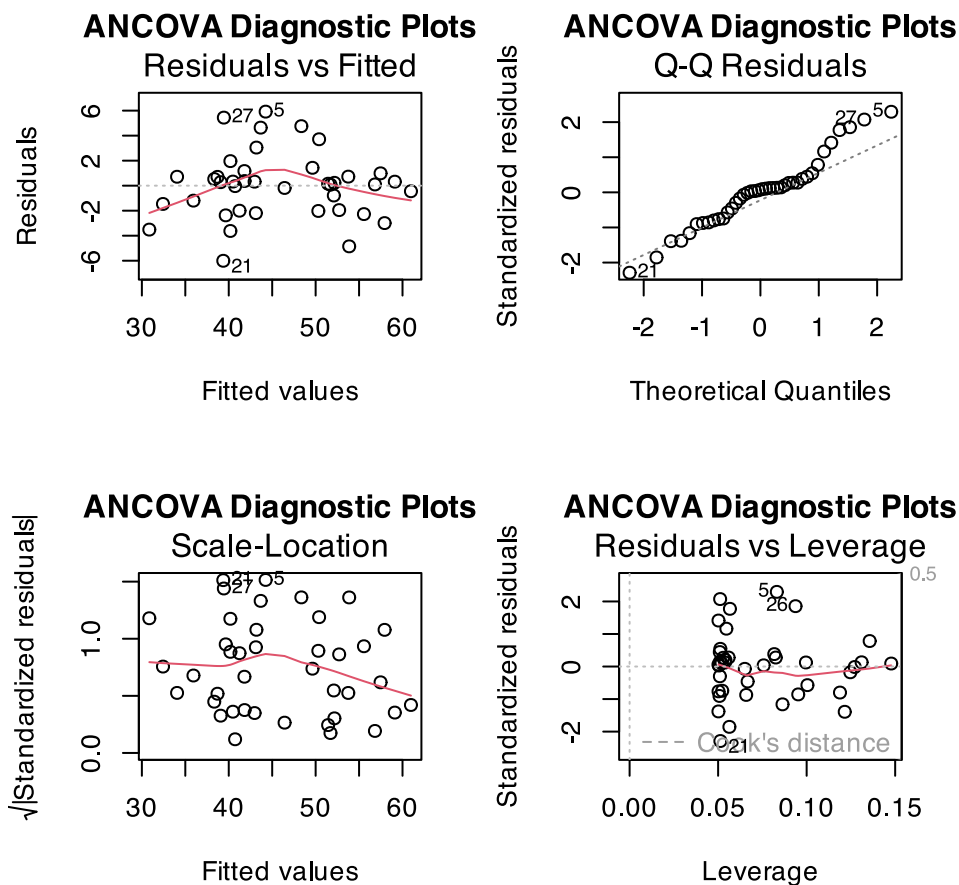
```
# View ANOVA table
Anova(cricket_ancova)
```

```
Anova Table (Type II tests)

Response: chirp_rate
              Sum Sq Df F value    Pr(>F)
temp          1348.81  1  185.90 0.0000000000000005296 ***
species       1368.34  1  188.59 0.0000000000000004236 ***
Residuals     268.46 37
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## Step 3: Check Model Assumptions

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(cricket_ancova, main = "ANCOVA Diagnostic Plots")
```



```
par(mfrow = c(1, 1))
```

## Step 4: Calculate Adjusted Means

ANCOVA compares adjusted means - what each group's mean would be at the overall mean of the covariate.

```
# Calculate adjusted means using emmeans
cricket_adjusted_means <- emmeans(cricket_ancova, "species")

# Convert to dataframe for plotting
cricket_adj_means_df <- as.data.frame(cricket_adjusted_means)
cricket_adj_means_df
```

| species          | emmean   | SE        | df | lower.CL | upper.CL |
|------------------|----------|-----------|----|----------|----------|
| 0. exclamationis | 51.70513 | 0.6049702 | 37 | 50.47934 | 52.93091 |
| 0. niveus        | 39.90462 | 0.6049702 | 37 | 38.67883 | 41.13040 |

Confidence level used: 0.95

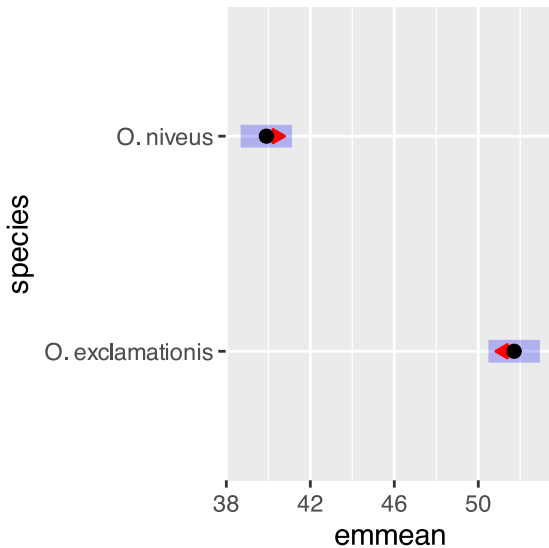
## Step 5: Pairwise Comparisons

```
# Pairwise comparisons of adjusted means
pairs(cricket_adjusted_means, adjust = "sidak")
```

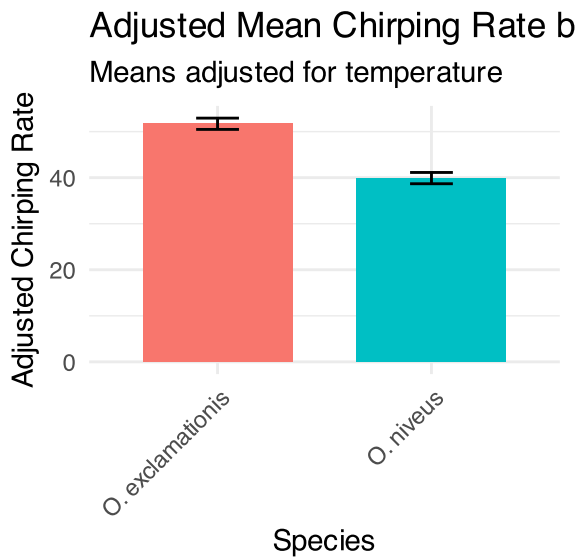
| contrast                     | estimate | SE    | df | t.ratio | p.value |
|------------------------------|----------|-------|----|---------|---------|
| 0. exclamationis - 0. niveus | 11.8     | 0.859 | 37 | 13.733  | <.0001  |

## Step 6: Visualize Results

```
# Plot adjusted means with confidence intervals
plot(cricket_adjusted_means, comparisons = TRUE)
```



```
# Bar chart of adjusted means
ggplot(cricket_adj_means_df, aes(x = species, y = emmean, fill = species)) +
  geom_bar(stat = "identity", width = 0.7) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2) +
  labs(title = "Adjusted Mean Chirping Rate by Species",
       subtitle = "Means adjusted for temperature",
       x = "Species",
       y = "Adjusted Chirping Rate") +
  theme_minimal() +
  theme(legend.position = "none",
       axis.text.x = element_text(angle = 45, hjust = 1))
```



## Part 2: Partridge Longevity Analysis

### Data Overview

We'll analyze the effect of mating strategy on male fruitfly longevity, using thorax length as a covariate.

```
# Load the partridge dataset
partridge <- read.csv("data/partridge.csv")

# Create better treatment names
partridge$treatment <- factor(partridge$TREATMEN,
                              levels = 1:5,
                              labels = c("No females",
                                           "One virgin female daily",
                                           "Eight virgin females daily",
                                           "One inseminated female daily",
                                           "Eight inseminated females daily"))

# View data structure
head(partridge)
```

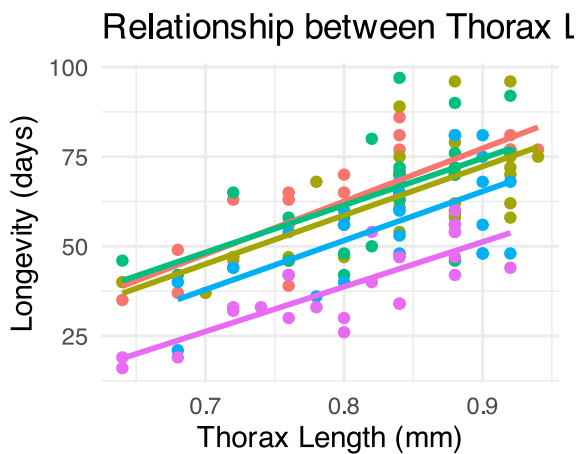
|   | PARTNERS | TYPE | TREATMEN | LONGEV | LLONGEV  | THORAX | RESID1     | PREDICT1 | RESID2      |
|---|----------|------|----------|--------|----------|--------|------------|----------|-------------|
| 1 | 8        | 0    | 1        | 35     | 1.544068 | 0.64   | -5.868456  | 40.86846 | -0.04743024 |
| 2 | 8        | 0    | 1        | 37     | 1.568202 | 0.68   | -9.301196  | 46.30120 | -0.07105067 |
| 3 | 8        | 0    | 1        | 49     | 1.690196 | 0.68   | 2.698804   | 46.30120 | 0.05094369  |
| 4 | 8        | 0    | 1        | 46     | 1.662758 | 0.72   | -5.733936  | 51.73394 | -0.02424867 |
| 5 | 8        | 0    | 1        | 63     | 1.799341 | 0.72   | 11.266064  | 51.73394 | 0.11233405  |
| 6 | 8        | 0    | 1        | 39     | 1.591065 | 0.76   | -18.166676 | 57.16668 | -0.14369601 |

|   | PREDICT2 | treatment  |
|---|----------|------------|
| 1 | 1.591498 | No females |
| 2 | 1.639252 | No females |
| 3 | 1.639252 | No females |
| 4 | 1.687007 | No females |
| 5 | 1.687007 | No females |
| 6 | 1.734761 | No females |

```
# Visualize the relationship between thorax length and longevity by treatment
ggplot(partridge, aes(x = THORAX, y = LONGEV, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship between Thorax Length and Longevity",
       x = "Thorax Length (mm)",
       y = "Longevity (days)",
       color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



nale daily — Eight virgin females daily — One ins

## Step 1: Test Homogeneity of Slopes

```
# Test for homogeneity of slopes
homo_slopes_model <- lm(LONGEV ~ THORAX * treatment, data = partridge)
Anova(homo_slopes_model, type = 3)
```

Anova Table (Type III tests)

Response: LONGEV

|                  | Sum Sq  | Df  | F value | Pr(>F)        |
|------------------|---------|-----|---------|---------------|
| (Intercept)      | 755.6   | 1   | 6.6320  | 0.01128 *     |
| THORAX           | 3486.3  | 1   | 30.5999 | 2.017e-07 *** |
| treatment        | 36.9    | 4   | 0.0810  | 0.98805       |
| THORAX:treatment | 42.5    | 4   | 0.0933  | 0.98441       |
| Residuals        | 13102.1 | 115 |         |               |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Step 2: Fit ANCOVA Model

```
# Fit the ANCOVA model (without interaction)
ancova_model <- lm(LONGEV ~ THORAX + treatment, data = partridge)
```

```
# Get more detailed summary
summary(ancova_model)
```

Call:

```
lm(formula = LONGEV ~ THORAX + treatment, data = partridge)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -26.189 | -6.599 | -0.989 | 6.408 | 30.244 |

Coefficients:

|                                          | Estimate | Std. Error | t value | Pr(> t ) |
|------------------------------------------|----------|------------|---------|----------|
| (Intercept)                              | -46.055  | 10.239     | -4.498  | 1.61e-05 |
| THORAX                                   | 135.819  | 12.439     | 10.919  | < 2e-16  |
| treatmentOne virgin female daily         | -3.929   | 2.997      | -1.311  | 0.192347 |
| treatmentEight virgin females daily      | -1.276   | 2.983      | -0.428  | 0.669517 |
| treatmentOne inseminated female daily    | -10.946  | 2.999      | -3.650  | 0.000391 |
| treatmentEight inseminated females daily | -23.879  | 2.973      | -8.031  | 7.83e-13 |

|                                          |     |
|------------------------------------------|-----|
| (Intercept)                              | *** |
| THORAX                                   | *** |
| treatmentOne virgin female daily         |     |
| treatmentEight virgin females daily      |     |
| treatmentOne inseminated female daily    | *** |
| treatmentEight inseminated females daily | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.51 on 119 degrees of freedom

Multiple R-squared: 0.6564, Adjusted R-squared: 0.6419

F-statistic: 45.46 on 5 and 119 DF, p-value: < 2.2e-16

```
# View ANOVA table
anova(ancova_model)
```

Analysis of Variance Table

Response: LONGEV

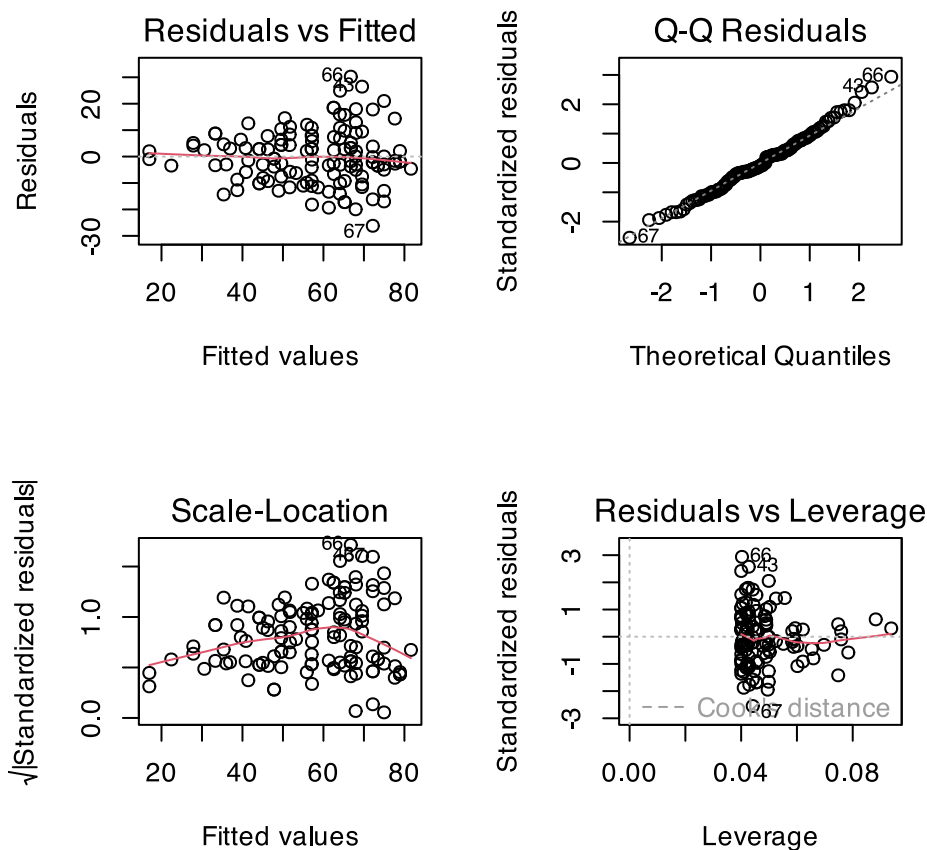
|           | Df  | Sum Sq  | Mean Sq | F value | Pr(>F)        |
|-----------|-----|---------|---------|---------|---------------|
| THORAX    | 1   | 15496.6 | 15496.6 | 140.293 | < 2.2e-16 *** |
| treatment | 4   | 9611.5  | 2402.9  | 21.753  | 1.719e-13 *** |
| Residuals | 119 | 13144.7 | 110.5   |         |               |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### Step 3: Check Assumptions

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(ancova_model)
```



## Step 4: Calculate Adjusted Means

```
# Get adjusted means using emmeans
adjusted_means <- emmeans(ancova_model, "treatment")
adjusted_means
```

| treatment                       | emmean | SE   | df  | lower.CL | upper.CL |
|---------------------------------|--------|------|-----|----------|----------|
| No females                      | 65.4   | 2.11 | 119 | 61.3     | 69.6     |
| One virgin female daily         | 61.5   | 2.11 | 119 | 57.3     | 65.7     |
| Eight virgin females daily      | 64.2   | 2.10 | 119 | 60.0     | 68.3     |
| One inseminated female daily    | 54.5   | 2.11 | 119 | 50.3     | 58.7     |
| Eight inseminated females daily | 41.6   | 2.12 | 119 | 37.4     | 45.8     |

Confidence level used: 0.95

## Step 5: Pairwise Comparisons

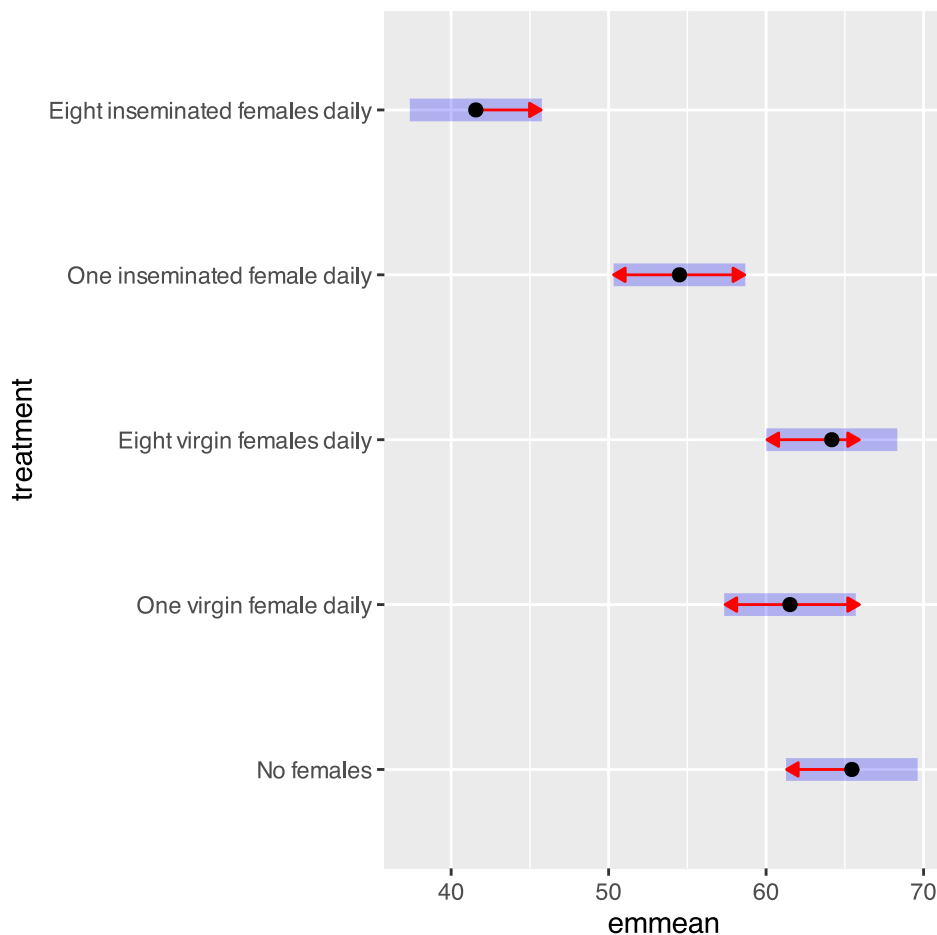
```
# Pairwise comparisons of adjusted means
pairs(adjusted_means, adjust = "tukey")
```

| contrast                                     | estimate | SE   |
|----------------------------------------------|----------|------|
| No females - One virgin female daily         | 3.93     | 3.00 |
| No females - Eight virgin females daily      | 1.28     | 2.98 |
| No females - One inseminated female daily    | 10.95    | 3.00 |
| No females - Eight inseminated females daily | 23.88    | 2.97 |

|                                                                |         |         |
|----------------------------------------------------------------|---------|---------|
| One virgin female daily - Eight virgin females daily           | -2.65   | 2.98    |
| One virgin female daily - One inseminated female daily         | 7.02    | 2.97    |
| One virgin female daily - Eight inseminated females daily      | 19.95   | 3.01    |
| Eight virgin females daily - One inseminated female daily      | 9.67    | 2.98    |
| Eight virgin females daily - Eight inseminated females daily   | 22.60   | 2.99    |
| One inseminated female daily - Eight inseminated females daily | 12.93   | 3.01    |
| df                                                             | t.ratio | p.value |
| 119                                                            | 1.311   | 0.6849  |
| 119                                                            | 0.428   | 0.9929  |
| 119                                                            | 3.650   | 0.0035  |
| 119                                                            | 8.031   | <.0001  |
| 119                                                            | -0.891  | 0.8996  |
| 119                                                            | 2.361   | 0.1336  |
| 119                                                            | 6.636   | <.0001  |
| 119                                                            | 3.249   | 0.0129  |
| 119                                                            | 7.560   | <.0001  |
| 119                                                            | 4.298   | 0.0003  |

P value adjustment: tukey method for comparing a family of 5 estimates

```
# Plot adjusted means with confidence intervals
plot(adjusted_means, comparisons = TRUE)
```



### Part 3: Example with Heterogeneous Slopes

Let's look at an example where slopes are NOT homogeneous using sea urchin data.

```

# Create simulated sea urchin data with heterogeneous slopes
set.seed(345)
n <- 72 # 24 urchins per group

# Create data frame
treatments <- rep(c("Initial", "Low Food", "High Food"), each = n/3)
volume <- c(
  runif(n/3, 10, 40), # Initial
  runif(n/3, 10, 40), # Low Food
  runif(n/3, 10, 40)  # High Food
)

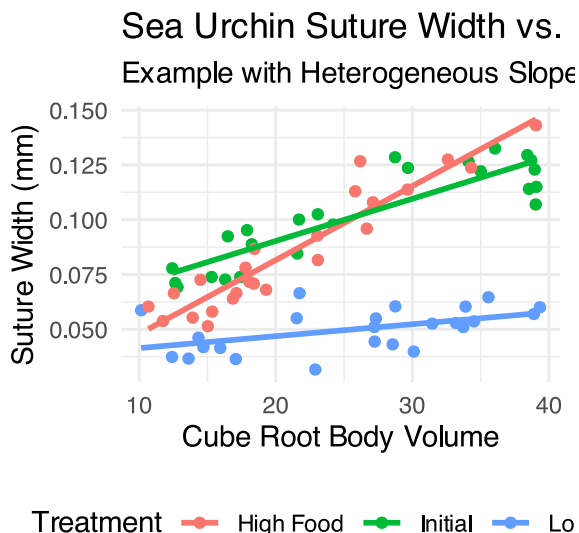
# Create suture width with different slopes for each treatment
suture_width <- ifelse(
  treatments == "Initial", 0.05 + 0.002 * volume,
  ifelse(
    treatments == "Low Food", 0.04 + 0.0005 * volume,
    0.02 + 0.003 * volume # High Food
  )
) + rnorm(n, 0, 0.01)

urchin_data <- data.frame(treatment = treatments, volume = volume, suture_width = suture_width)

# Plot the data with regression lines
ggplot(urchin_data, aes(x = volume, y = suture_width, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sea Urchin Suture Width vs. Volume",
        subtitle = "Example with Heterogeneous Slopes",
        x = "Cube Root Body Volume",
        y = "Suture Width (mm)",
        color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

`geom\_smooth()` using formula = 'y ~ x'



## Test for Homogeneity of Slopes

```
# Fit model with interaction
urchin_model <- lm(suture_width ~ volume * treatment, data = urchin_data)
Anova(urchin_model, type = 3)
```

Anova Table (Type III tests)

Response: suture\_width

|                  | Sum Sq    | Df | F value | Pr(>F)        |
|------------------|-----------|----|---------|---------------|
| (Intercept)      | 0.0005253 | 1  | 5.91    | 0.01778 *     |
| volume           | 0.0151663 | 1  | 170.64  | < 2.2e-16 *** |
| treatment        | 0.0020070 | 2  | 11.29   | 6.064e-05 *** |
| volume:treatment | 0.0062129 | 2  | 34.95   | 4.453e-11 *** |
| Residuals        | 0.0058662 | 66 |         |               |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**Result:** With  $p < 0.05$ , we have heterogeneous slopes! Standard ANCOVA would be inappropriate here.

## What to do with Heterogeneous Slopes

When slopes are not homogeneous, you have several options:

```
# Option: Analyze groups separately
initial_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "Initial"))
low_food_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "Low Food"))
high_food_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "High Food"))

# Summary for each group
initial_model
```

Call:

```
lm(formula = suture_width ~ volume, data = filter(urchin_data,
  treatment == "Initial"))
```

Coefficients:

|             |          |
|-------------|----------|
| (Intercept) | volume   |
| 0.051785    | 0.001926 |

low\_food\_model

Call:

```
lm(formula = suture_width ~ volume, data = filter(urchin_data,
  treatment == "Low Food"))
```

Coefficients:

|             |           |
|-------------|-----------|
| (Intercept) | volume    |
| 0.0359532   | 0.0005453 |

```
high_food_model
```

Call:

```
lm(formula = suture_width ~ volume, data = filter(urchin_data,  
  treatment == "High Food"))
```

Coefficients:

|             |          |
|-------------|----------|
| (Intercept) | volume   |
| 0.014077    | 0.003376 |

## Summary Checklist for ANCOVA

When conducting ANCOVA, always follow these steps:

### 💡 ANCOVA Checklist

1. **Visualize your data** - plot response vs covariate, colored by groups
2. **Test homogeneity of slopes** - fit model with interaction term
  - If  $p > 0.05$ : proceed with ANCOVA
  - If  $p < 0.05$ : use alternative approaches
3. **Fit ANCOVA model** - response ~ covariate + factor
4. **Check assumptions** - use diagnostic plots
5. **Interpret results** - focus on adjusted means, not raw means
6. **Conduct post-hoc tests** - pairwise comparisons if needed
7. **Visualize results** - show adjusted means with confidence intervals

## Key Points to Remember

- **ANCOVA increases power** by accounting for covariate variation
- **Adjusted means** are what we compare, not raw group means
- **Homogeneity of slopes** is the most critical assumption
- **Parallel lines** in your plot suggest homogeneous slopes
- **Non-parallel lines** indicate heterogeneous slopes - use alternative methods

### ! Key Points from ANCOVA Analysis

1. **Test homogeneity of slopes first** - this is the most critical assumption
2. **ANCOVA compares adjusted means** at the mean value of the covariate
3. **Increases statistical power** by removing variation due to the covariate
4. **Choose appropriate methods** based on whether slopes are homogeneous
5. **Visualize your results** clearly showing the relationship between variables
6. **Check all assumptions** using diagnostic plots
7. **Interpret in biological context** - what do the adjusted means tell us?

Remember: The covariate should be measured independently of the treatment and should not be affected by the treatment itself!